EXTENSION OF SCALARS OF AN INJECTIVE MODULE ALONG A POLYNOMIAL RING HAS INJECTIVE AMPLITUDE IN TWO DEGREES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-dualizing/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The statement that a direct sum of injective modules over a Noetherian ring is injective. Invoked inside the counted proof (source lines 608-609). Makes the extended module injective as a module over the base ring.
- The statement that the module of maps from a ring into an injective module is injective. Invoked at the end of the counted proof (source lines 610-611). Makes the middle term of the displayed sequence injective.

Source reference. Original lemma lemma-hom-injective: sources/stacks-dualizing/pinned-original-source.tex, source lines 433–433. Navigation pointer only: the counted text refers to the Stacks item lemma-hom-injective, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-sum-injective-modules: sources/stacks-dualizing/pinned-original-source.tex, source lines 509–509. Navigation pointer only: the counted text refers to the Stacks item lemma-sum-injective-modules, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring. Let E be an injective A-module. Then $E \otimes_A A[x]$ has injective-amplitude $[0, 1]$ as an object of $D(A[x])$. In particular, $E \otimes_A A[x]$ has finite injective dimension as an $A[x]$-module.*

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let us write $E[x] = E \otimes_A A[x]$. Consider the short exact sequence of $A[x]$-modules

$$0 \to E[x] \to \mathrm{Hom}_A(A[x], E[x]) \to \mathrm{Hom}_A(A[x], E[x]) \to 0$$

where the first map sends $p \in E[x]$ to $f \mapsto fp$ and the second map sends φ to $f \mapsto \varphi(xf) - x\varphi(f)$. The second map is surjective because $\mathrm{Hom}_A(A[x], E[x]) = \prod_{n \geq 0} E[x]$ as an abelian group and the map sends (e_n) to $(e_{n+1} - xe_n)$ which is surjective. As an A-module we have $E[x] \cong \bigoplus_{n \geq 0} E$ which is injective by Lemma lemma-sum-injective-modules. Hence the $A[x]$-module $\mathrm{Hom}_A(A[x], E[x])$ is injective by Lemma lemma-hom-injective and the proof is complete. $\square$